

DAY-4

Student Name: Saloni Kumari

Branch: MCA

Subject Name: LB 3

UID: 25MCA20087

Section/Group: MCA-3

Date of Performance: 10-07-2026

Q1. Imagine a building or street network represented as a graph:

- **Nodes (vertices):** Intersections, rooms, or key areas.
- **Edges:** Paths, corridors, or streets connecting the nodes.

Install the minimum number of surveillance cameras such that every connection (edge) is monitored, meaning at least one of its endpoints (nodes) has a camera.

This is the Minimum Vertex Cover problem.

Code:

```
graph = {
```

```
    0: [1, 2],
```

```
    1: [0, 2],
```

```
    2: [0, 1, 3],
```

```
    3: [2]
```

```
}
```

```
visited = set()
```

```
print("Vertices with Cameras:")
```

```
for u in graph:
```

if u not in visited:

for v in graph[u]:

if v not in visited:

print(u)

visited.add(u)

visited.add(v)

break

Output:

```
PS C:\Users\salon\OneDrive\Desktop\EST> python lg_cp1.py
Vertices with Cameras:
0
2
PS C:\Users\salon\OneDrive\Desktop\EST>
```

Q2. Verify whether a feasible exam timetable exists such that no student has overlapping exams.

This is solved using **Graph Coloring**.

Code:

```
graph = {
    0: [1, 2],
    1: [0, 2],
    2: [0, 1, 3],
    3: [2]
}
```

```
colors = {}
```

```
def is_safe(node, color):
    for neighbor in graph[node]:
        if colors.get(neighbor) == color:
```

```
    return False  
    return True
```

```
for node in graph:  
    color = 1  
    while not is_safe(node, color):  
        color += 1  
    colors[node] = color  
  
print("Exam Time Slots:")  
for node in colors:  
    print("Exam", node, "-> Slot", colors[node])
```

Output:

```
PS C:\Users\salon\OneDrive\Desktop\EST> python lg_cp1.py  
Exam Time Slots:  
Exam 0 -> Slot 1  
Exam 1 -> Slot 2  
Exam 2 -> Slot 3  
Exam 3 -> Slot 1  
PS C:\Users\salon\OneDrive\Desktop\EST>
```



Saloni Singh 
@saloniraj8757

[Complete your profile](#)
Add your missing details →
This data will be helpful to auto-fill your job applications

Personal Information

 saloniraj8757@gmail.com
 Add your mobile number
 India

My Badges


Python

My Resume [+ Add Resume](#)
Add your resume here

My Certifications
You have not earned any certificates yet. [Get Certified](#)



UNIVERSITY INSTITUTE *of*
COMPUTING



<https://www.hackerrank.com/profile/saloniraj8757>